

Lab1: Introduction to XAMPP, HTML, and CSS

Objective

- Install and configure XAMPP
- Create and deploy a simple webpage (HTML document)
- Customize a webpage style with CSS

Resource

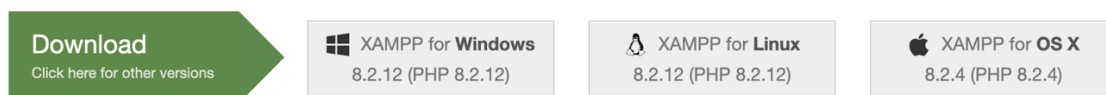
- XAMPP is an easy to install Apache server distribution containing MariaDB, PHP, and Perl. We can make our web and database server ready by simple click "next".
- VS Code is the recommended IDE. IDE (integrated development environment) can help us to have a better development experience. There are many other choices, such as Sublime Text and WebStorm. You can choose your favourite one.

Tips

- These labs are expected to take more than the allocated 2 hours. You should try the tasks before the lab. Practice makes code perfect.
- Before the lab, you could post your questions in LMO forum, and you are welcome to offer solutions to the questions in LMO.
- During the lab, you could discuss with TAs for your questions. Face to face discussion will have a relatively high efficiency.

A. Install XAMPP

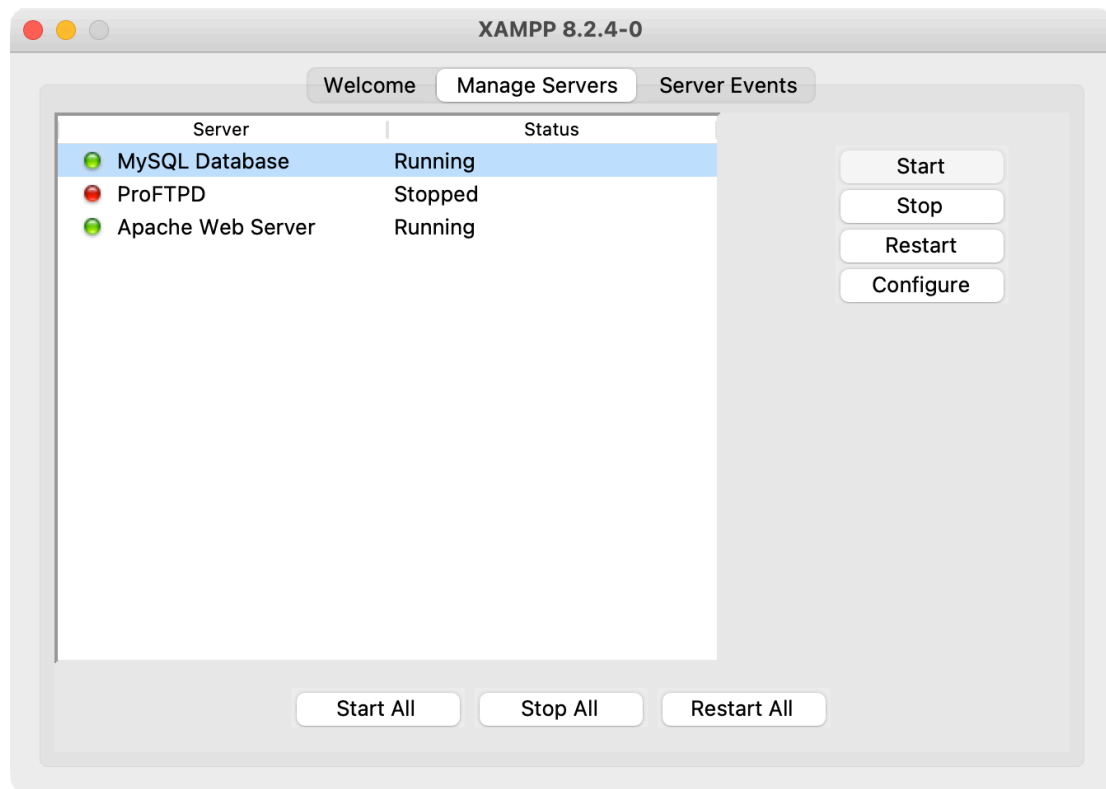
1. Download the XAMPP at <https://www.apachefriends.org/index.html>



2. Please choose the appropriate version based on your operating system. If you have already installed older versions of XAMPP, ensure that the PHP version is 7.3 or higher.

3. Once XAMPP is installed, you can launch it by clicking the "manager-osx" in Launchpad on macOS or searching for "XAMPP" in the Start menu on Windows. It will look like:

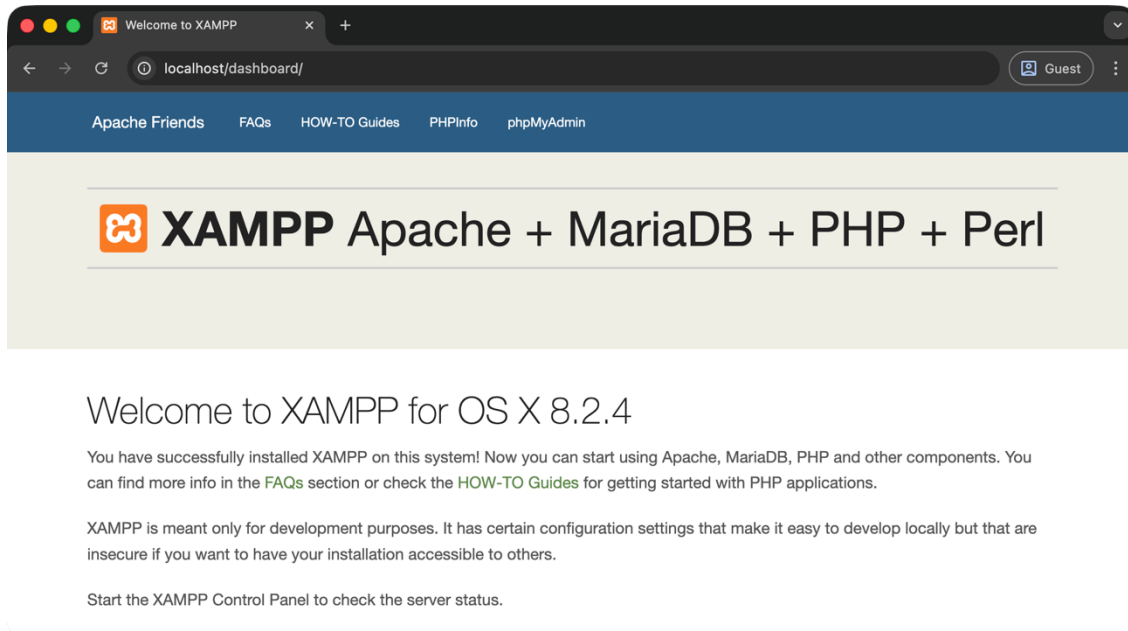
(a) macOS:



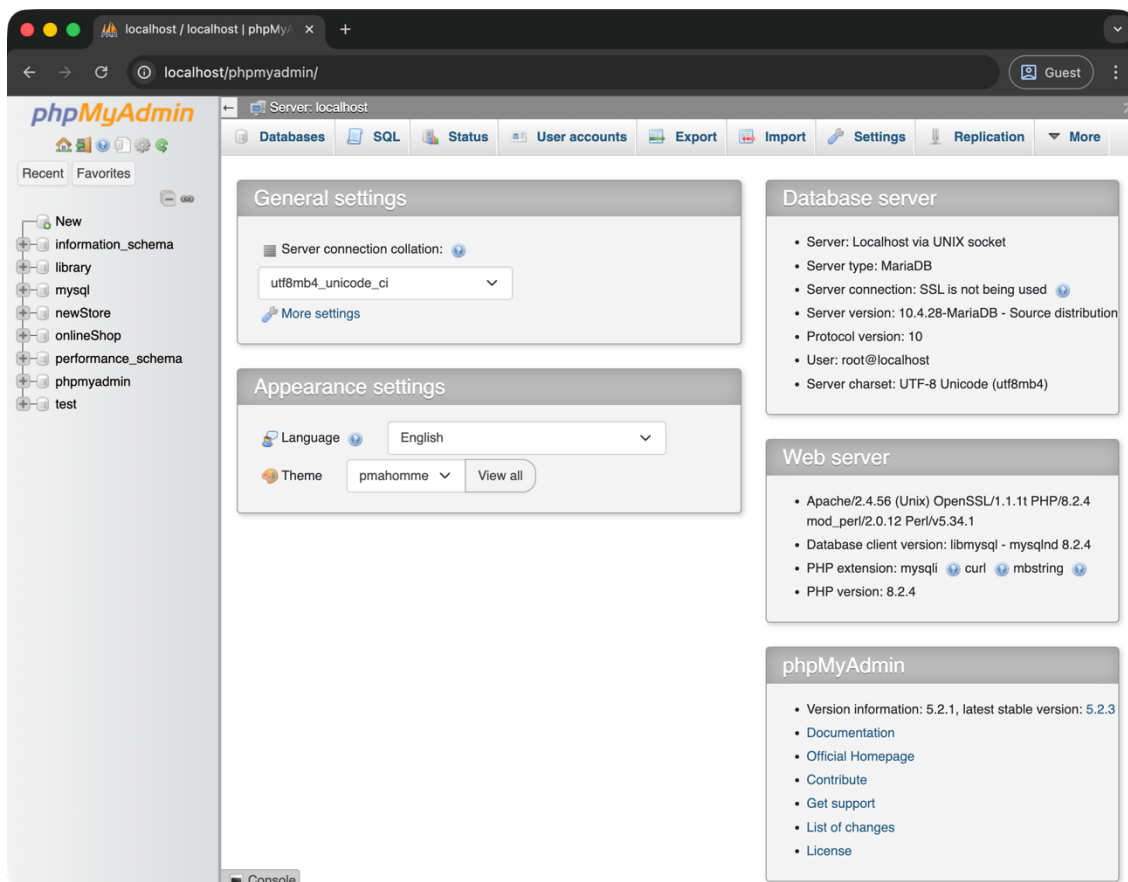
(b) Windows:



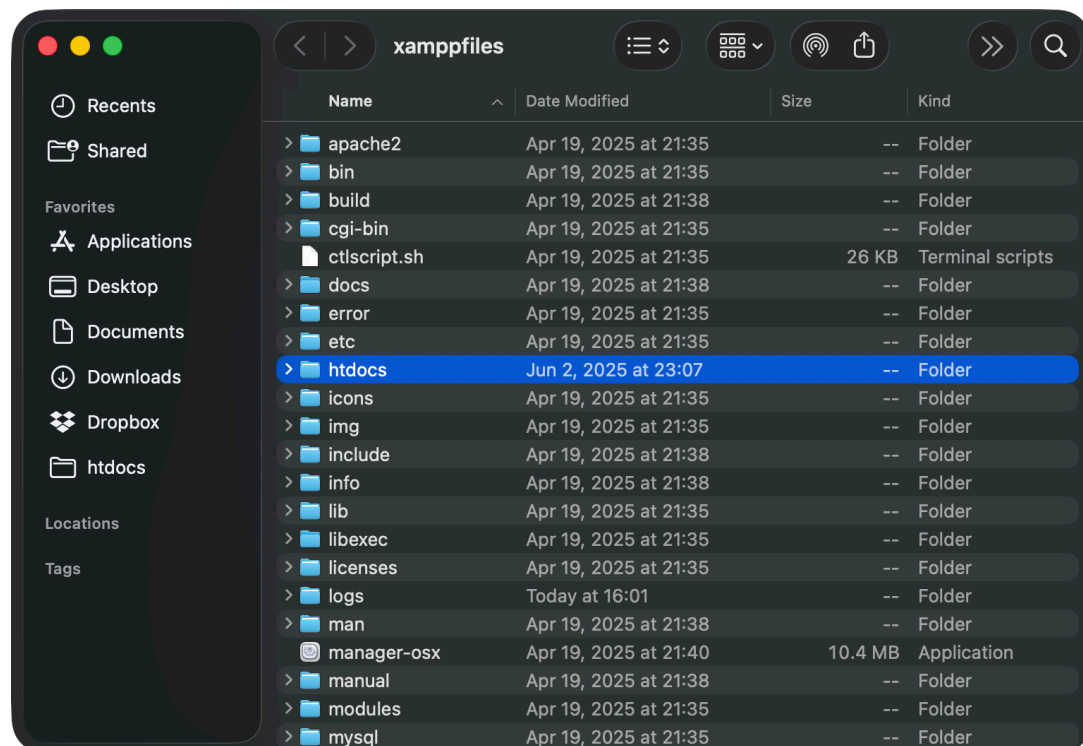
4. Open a web browser and enter "localhost" in the address field. You should see the default page of Apache server:



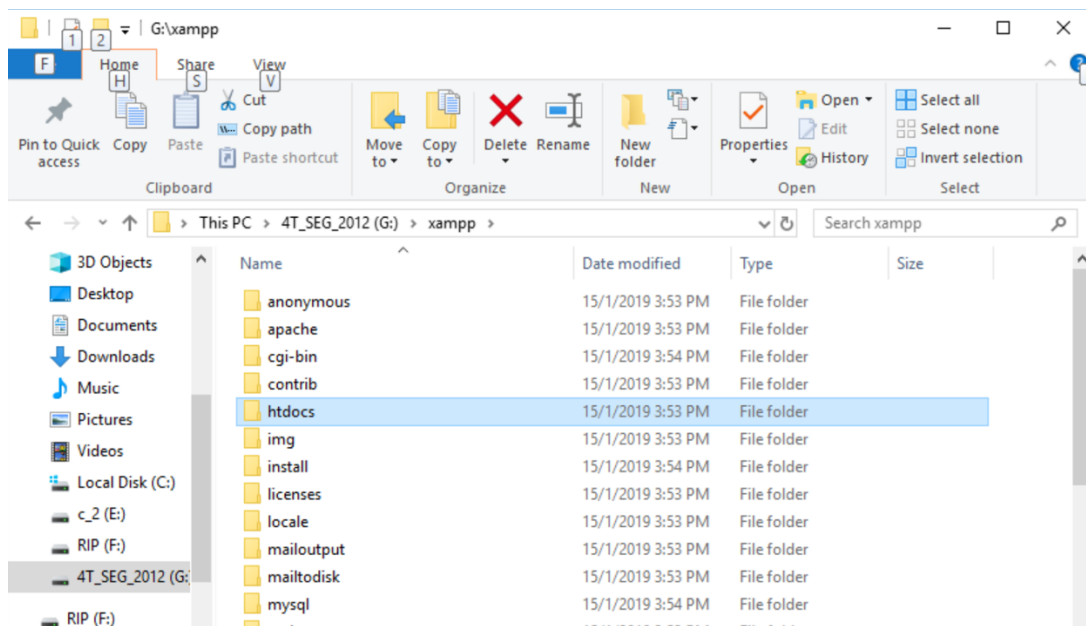
5. By click on the "phpMyAdmin" button, you should see the MySQL server:



6. The default root directory of Apache server is the "htdocs" folder. You can find them as:
- (a) macOS (click "Welcome" -> "Open Application Folder"):



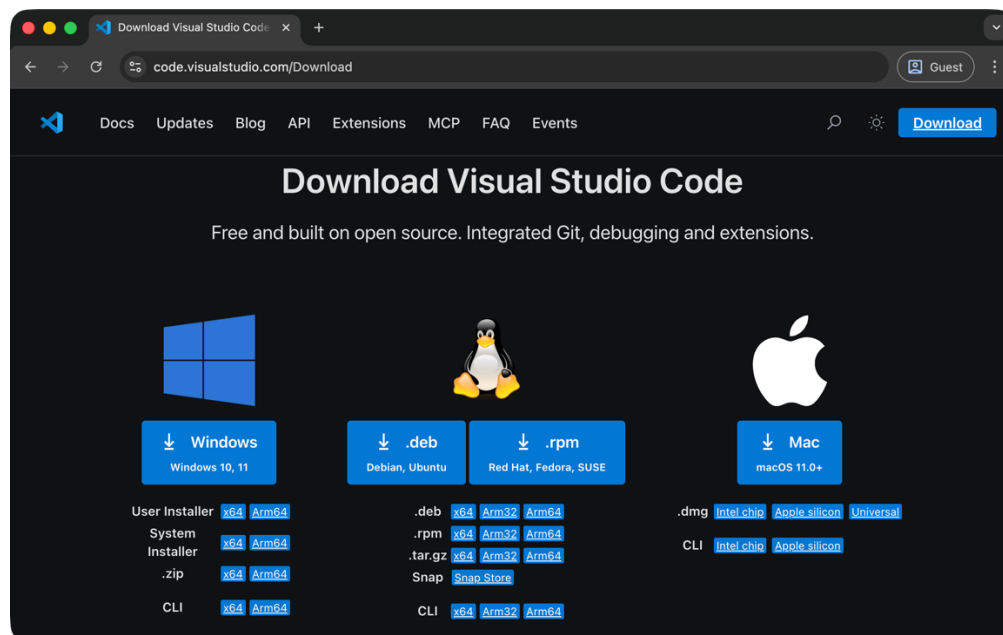
(b) Windows (depending on where you installed XAMPP):



7. You can add your web contents in this folder, they will be shown in your web. You may create a shortcut to "htdocs" for easy access.

B. Install VS Code (optional)

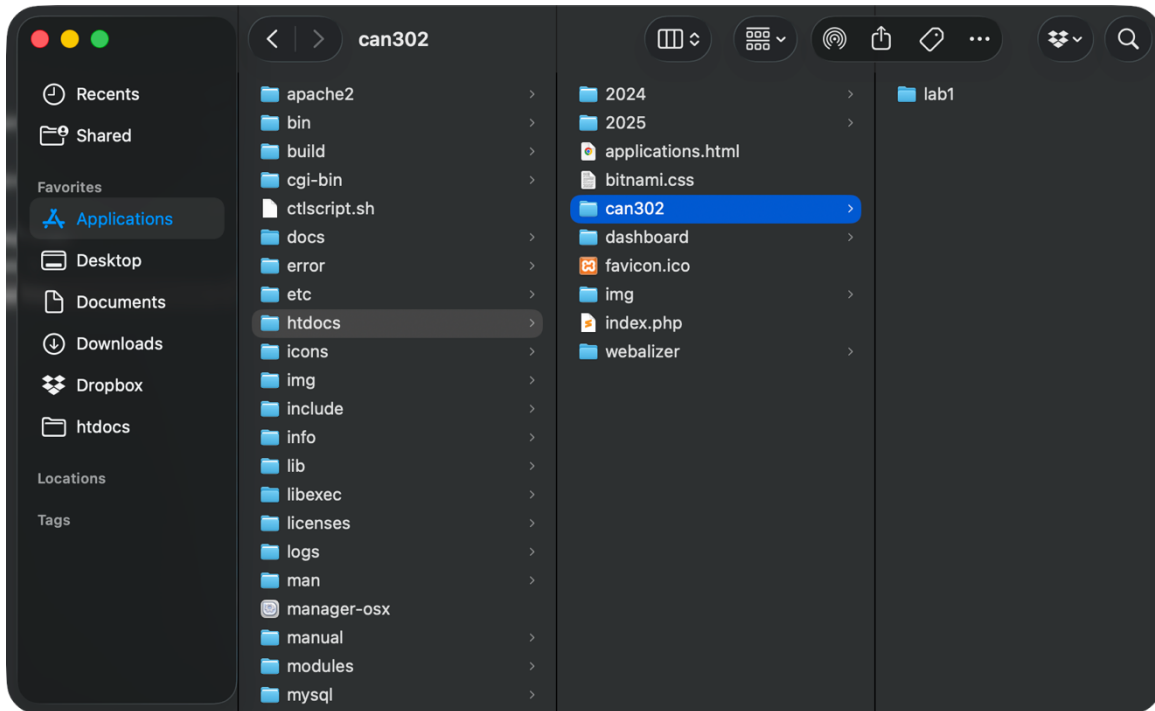
1. Download the VS Code at <https://code.visualstudio.com/download>



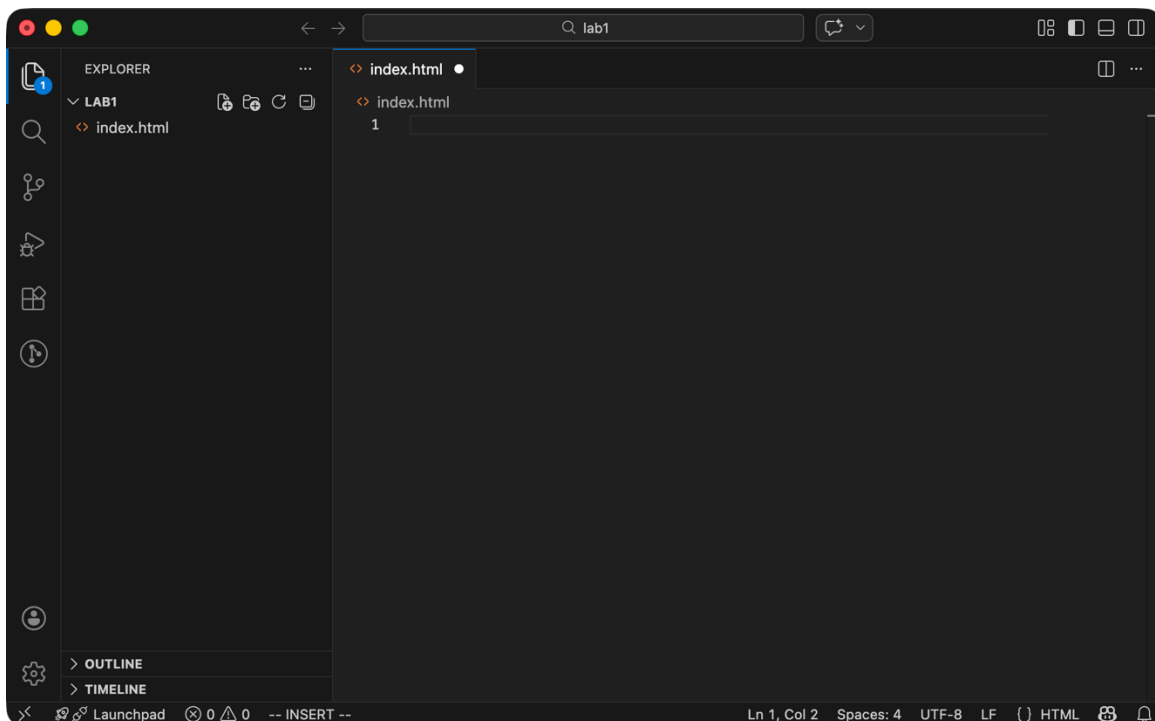
2. Please choose the appropriate version based on your operating system.

C. Build a Webpage

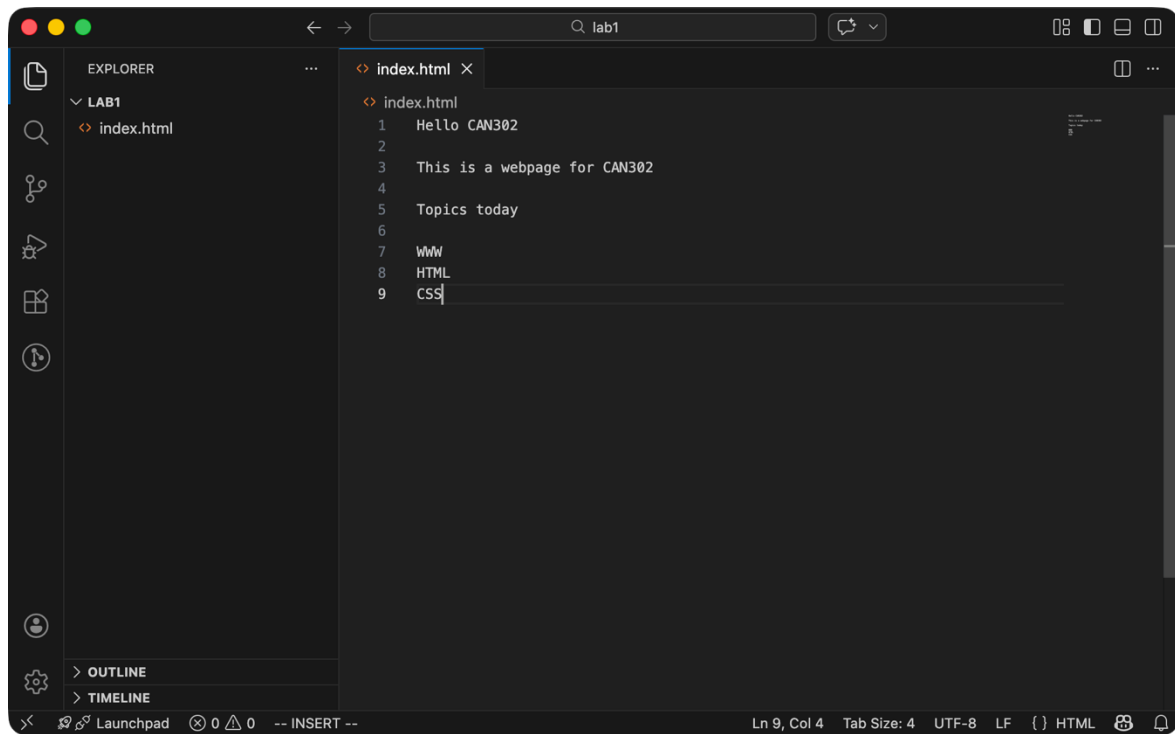
1. Create a folder "can302" under "htdocs", and then create a subfolder "lab1" under "can 302"
(The following screenshot is in macOS, but same operations can be applied to Windows):



2. Use VS Code to open the "lab1" folder and create a new file named "index.html":



3. Input text in "index.html" as follows and save it:



The screenshot shows a code editor with a dark theme. The Explorer panel on the left shows a folder named 'LAB1' containing a file 'index.html'. The main editor area displays the content of 'index.html' with line numbers 1 through 9. The text is as follows:

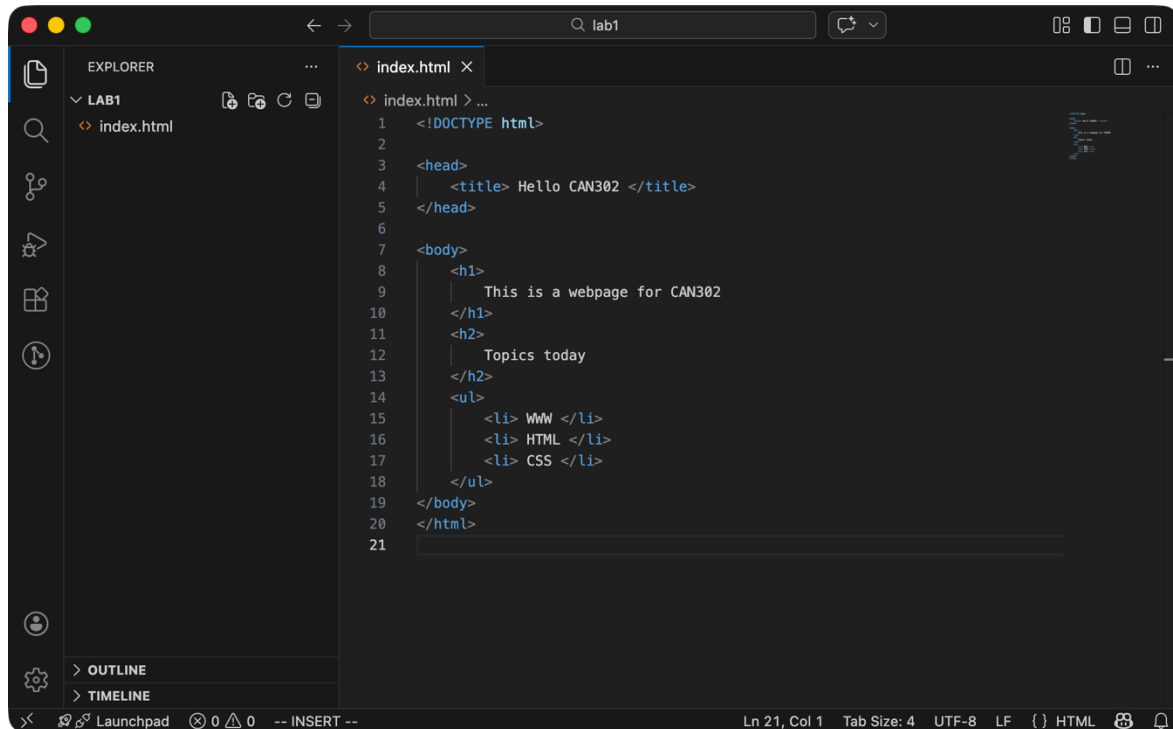
```
<?xml version="1.0" encoding="UTF-8" ?>
1  Hello CAN302
2
3  This is a webpage for CAN302
4
5  Topics today
6
7  WWW
8  HTML
9  CSS
```

The status bar at the bottom indicates 'Ln 9, Col 4', 'Tab Size: 4', 'UTF-8', 'LF', and 'HTML'.

4. Access the webpage in your browser via URL: <http://localhost/can302/lab1>



5. Add HTML tags to the document as below, and save file after finish:



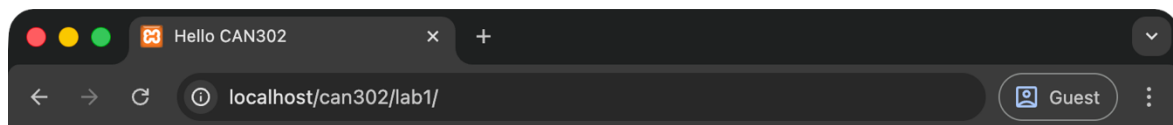
The screenshot shows a code editor with a dark theme. The Explorer panel on the left shows a file named 'index.html' under a folder named 'LAB1'. The main editor area displays the following HTML code:

```
<!DOCTYPE html>

<head>
  <title> Hello CAN302 </title>
</head>

<body>
  <h1>
    This is a webpage for CAN302
  </h1>
  <h2>
    Topics today
  </h2>
  <ul>
    <li> WWW </li>
    <li> HTML </li>
    <li> CSS </li>
  </ul>
</body>
</html>
```

6. Reload the webpage:



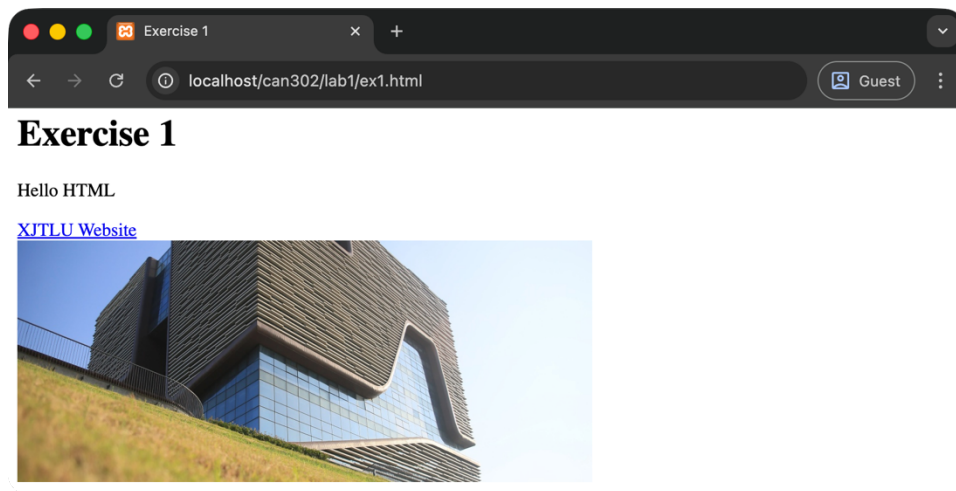
This is a webpage for CAN302

Topics today

- WWW
- HTML
- CSS

D. Hand-on Exercises

1. Create a HTML file that will generate the following outlook:



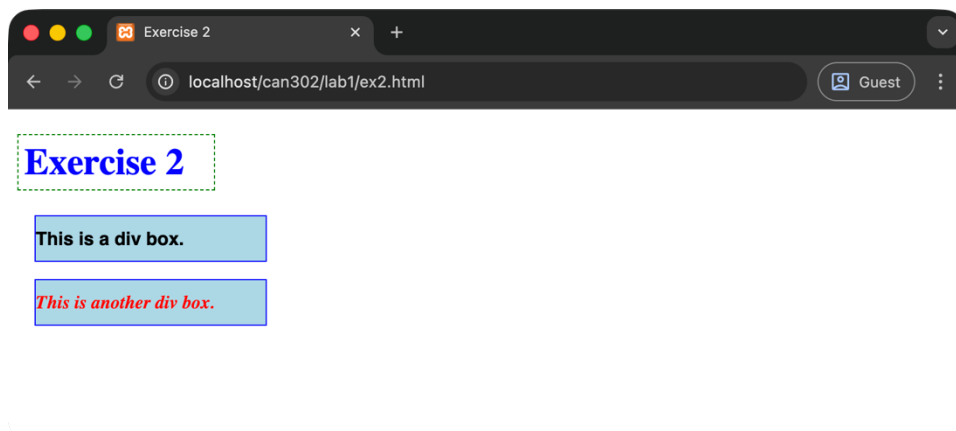
(a) Requirements:

- Exercise 1 (Heading 1)
- Hello HTML (paragraph)
- Hyperlink links to <https://www.xjtlu.edu.cn/en>
- Image URL: <https://www.xjtlu.edu.cn/wp-content/uploads/2022/10/central-building.1f5be76d-1-1.jpg>
- Image attributes: alt="XJTLU", width=500px, height=210px

(b) Useful references:

- Hyperlink https://www.w3schools.com/tags/tag_a.asp
- Image https://www.w3schools.com/tags/tag_img.asp

2. Use HTML and CSS to create a webpage similar to the one below:



Summary

We have achieved the following things in this lab session:

- Install and setup a web server (XAMPP)
- Deploy a simple HTML webpage
- Use CSS to customize the style of a webpage